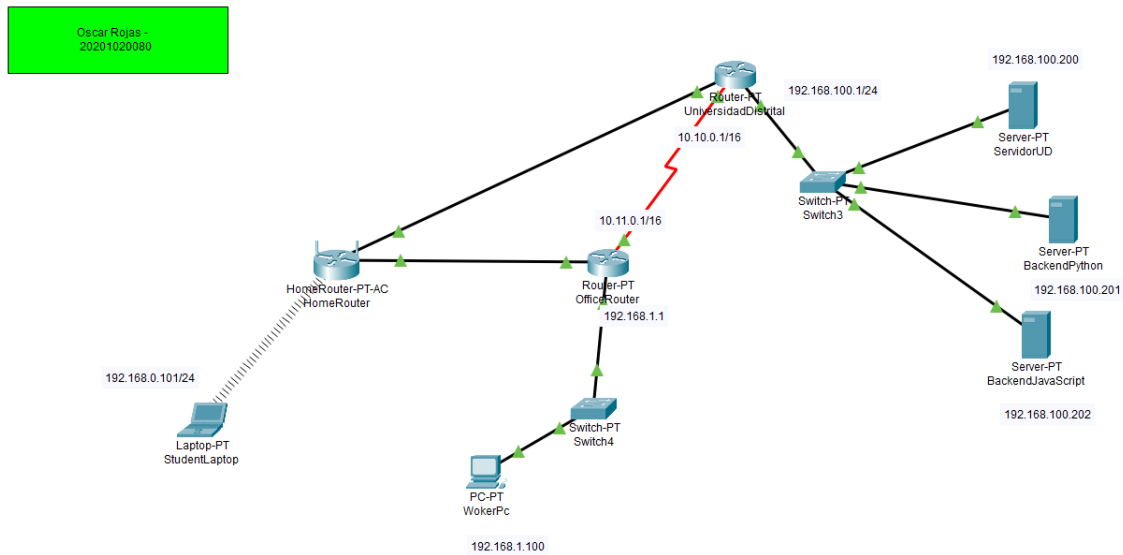


Topology



To design the network depicted, I aimed to establish an interconnected system with clear separation of networks based on their functions. The HomeRouter was set up to manage the student's personal devices in the 192.168.0.0/24 subnet, ensuring a private and secure environment. The OfficeRouter was configured for work-related devices in the 192.168.1.0/24 subnet, isolating professional operations. Both routers were interconnected via the 10.11.0.0/16 network, acting as the backbone for communication between these private networks and the central UniversidadDistrital Router, which also connects to the 192.168.100.0/24 subnet where servers are hosted. The servers were divided into separate nodes for backend services (Python, JavaScript) and institutional resources (ServidorUD) to streamline service accessibility. Each router's routing table was configured to allow seamless communication across subnets, and NAT (if needed) ensures external Internet access. The design prioritizes logical segmentation, scalability, and efficient traffic management.

Using the Simulation option in Packet Tracer, I analyzed how data flows through the network across the OSI model layers. This feature allowed me to visualize the encapsulation and de-encapsulation processes as packets traversed the network. By observing the flow of packets between devices, I verified proper communication within individual subnets, such as 192.168.0.0/24 for the student network and 192.168.100.0/24 for the servers. However, due to time constraints, I was unable to fully configure routing tables or implement protocols necessary for interconnection

between all private networks. As a result, devices in one private subnet could not reach devices in another. This limitation highlighted the importance of configuring inter-router communication and ensuring accurate routing across all subnets to establish seamless end-to-end connectivity.